

CAREER ARTIFACT · EVIDENCE-GROUNDED · 2026

ARIF FAZIL

SUBSURFACE · SYSTEMS · INTELLIGENCE

Thirteen years of exploration geoscience across the Malaysian and Southeast Asian subsurface. One continuing question, asked of every dataset and every model: what does reality actually support?

PETRONAS CARIGALI · EXPLORATION GEOSCIENCE
KUALA LUMPUR, MALAYSIA · 2013 – 2026

REALITY > EVERYTHING
DITEMPA BUKAN DIBERI

Who this is, and why it matters.

A subsurface geoscientist who learned to connect geology, uncertainty, evidence, human judgment and machine intelligence — not a geologist who used AI, and not an AI person who read a geology book.

EARTH

Thirteen years reading the subsurface across the **Malay Basin** and **offshore Sabah** — seismic interpretation, structural and sequence stratigraphy, petroleum systems, prospect maturation. Work that has carried prospects from regional screening to **drill-or-drop**, including first-of-trend plays and honest close-outs.

DECISIONS

Depth in the part of exploration that is not about rock but about **judgment under incomplete evidence**: risking, volumetrics, value-of-information, and knowing when the data says **stop**. Has held subsurface accountability across a **portfolio of operated platforms**, not a single field.

SYSTEMS

Turned that discipline into software. Author and architect of **arifOS** — a governance layer that checks a proposed AI action against constitutional constraints **before it executes**. An independent initiative; the same instinct that separates a checked interpretation from a persuasive one.

13

YEARS EXPLORATION
GEOSCIENCE

2

BASIN THEATRES —
MALAY & SABAH

4+

WELLS NAMED IN THE
PUBLIC RECORD

1

INDEPENDENT KERNEL —
PUBLISHED, LIVE

The question that organizes the work is not “*what do we know?*” — it is “**what does reality actually support, and how much of that can I defend?**”

2013 → 2026

A PETRONAS scholarship in 2009, a dual degree in geology and economics, and thirteen years inside one of the region's most consequential institutions. Every milestone below is anchored to a documented record.

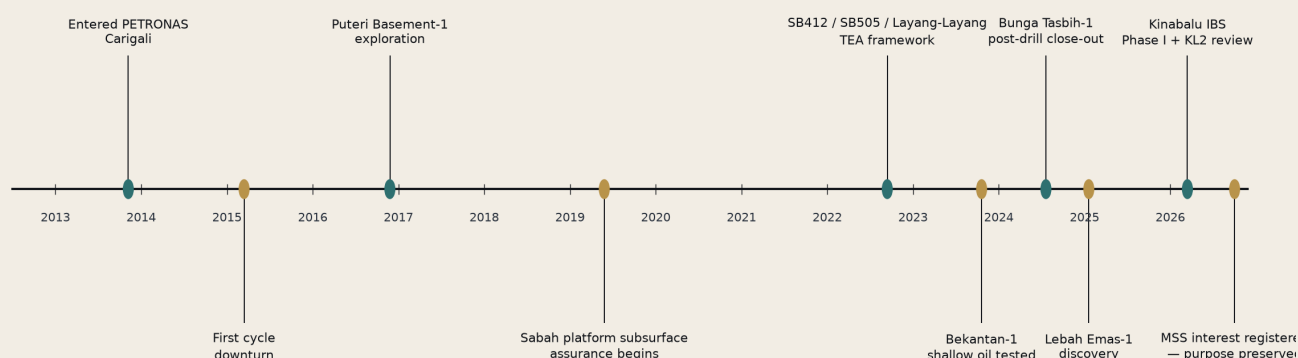


FIGURE 02 — Career chronology. Dates from the documented career record (CV v2, Sept 2026; career form, 29 Sept 2026). No item is dated from memory.

WHAT HAS HELD

A consistent technical spine — **read the data, hold the interpretation, carry it to a decision.** Across regime changes, price cycles and reorganisations, the constant is depth in one theatre rather than breadth across many.

WHAT HAS GROWN

From prospect-level work to **portfolio-scale accountability**, then to **systems**: transferable reasoning, governance of data, and finally a discipline applied to machine intelligence itself.

Two theatres, one margin.

The Malay Basin and offshore NW Sabah are not two separate stories — they are different expressions of the same regional tectonic machine. Reading that connection is a core part of the technical contribution.

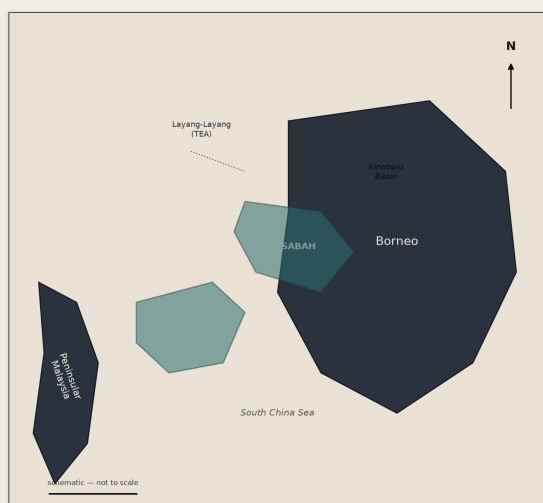


FIGURE 03 — Regional position, schematic only. No coordinates, no licence geometry, no confidential data. Shapes are representative.
TRUTH CLASS: SYNTHETIC EXPLANATORY FIGURE.

MALAY BASIN

Mature, well-drilled, and therefore unforgiving — the place where a model is tested fastest and where superficial understanding is least survivable.

Bekantan-1, Lebah Emas-1, Bunga Tasbih-1 and the basement play concept sit here.

OFFSHORE SABAH

Frontier and deepwater — sparse data, high consequence, structurally complex. The **Kinabalu Basin** integrated study, the **SB412 / SB505 / Layang-Layang** evaluation area, the inherited dry-hole corpus, and platform subsurface assurance all belong to this theatre.

Basin names and well names are used as published identifiers only. No reserve, resource, or economic figure is stated anywhere in this document.

Capability, evidence-tagged.

Each capability is tagged by the strongest evidence that supports it — not by self-assessment.

Proven means delivered and documented in the project record; **applied** means integrated into workflow; **evidenced** means mentored and handed over.

SUBSURFACE CAPABILITY

each row evidence-tagged to the project record



FIGURE 04 — Capability matrix. Tiers from the CV v2 record (Sept 2026) and the ExxonMobil CV competency list (Apr 2026). Tags describe evidence strength, not claimed mastery.

INTERPRETATION ENVIRONMENT

Daily practice in **DecisionSpace / DSG** — seismic interpretation, well-log and pressure integration, database audit and recovery after first-hand experience of an unrecoverable interpretation-data loss.

COMPUTATION

Probabilistic thinking over dogma: Bayesian volumetrics, posterior preservation, correlation guards — the same maths whether it is applied to a prospect or a model.

Discovery, and the discipline beside it.

LEBAH EMAS-1 · PM6-12 · MALAY BASIN · 2025

An eleven-reservoir discovery.

SENIOR EXPLORATION GEOSCIENTIST · PROSPECT-TO-WELL LEAD

THE PROBLEM

Mature Malay Basin acreage where the standing view had already mapped the productive fairway. The question was whether a stacked-sand architecture — the kind modelled on the Gulf of Thailand play — could be demonstrated at the prospect location.

THE EARTH SIGNAL

Seismic amplitude and motif analysis combined with well-tie calibration pointed to a multi-storey clastic system with more hydrocarbon-bearing intervals than the initial model had allowed for.

ROLE & DECISION

Led the integrated interpretation — seismic, petrophysics, structural — and retained authorship across the evaluation cycle. Recommended the well; supported the regulatory and authorisation path to spud.

WHAT FOLLOWED

The well carried **eleven hydrocarbon-bearing reservoirs**, confirming the stacked-sand architecture and adding a new reference case for the play going forward. No volumes or economic figures are stated here — those are not publication-safe, and this document does not manufacture them.

WHAT WAS PROVEN

Multi-reservoir prospect maturation, and the ability to hold an interpretation all the way to a live decision.

BEKANTAN-1 · PM304 · MALAY BASIN · 2024

Testing the depth model the portfolio had closed.

EXPLORATION GEOSCIENTIST · PROSPECT GENERATION

THE PROBLEM

The shallow section of the play had been treated as below the working oil horizon — effectively a closed question. Testing it meant arguing against the standing portfolio view.

THE EARTH SIGNAL

A four-way anticline trap in the shallow Miocene section became interpretable once the charge model was rebuilt. Basin modelling — not optimism — is what made the case.

ROLE & DECISION

Identified the trap, ran the charge risk assessment, prepared the prospect risking matrix, and recommended the well location.

WHAT FOLLOWED

The well established the **shallowest oil accumulation tested in the Malay Basin** at the time, and reframed the depth-horizon model for the play — a basin-model revision that begins with a well, not a memo.

WHAT WAS PROVEN

Willingness to test a basin model the portfolio had already closed — the harder half of exploration.

Building the framework where none existed.

KINABALU BASIN INTEGRATED STUDY · PHASE I · OFFSHORE SABAH

Eighty-eight wells into one defensible model.

INTEGRATED STUDY · REGIONAL FRAMEWORK AUTHOR

THE PROBLEM

Sparse, conflicting datasets across a structurally complex frontier basin. Regional questions had no shared reference framework — every team was reading from a different map.

THE EARTH SIGNAL

Compiling an **88-well database** exposed a seismic-motif signature that mapped a new play indicator. It also surfaced contested biostratigraphic questions that could not be left unresolved if the framework was to hold.

ROLE & DECISION

Compiled the database, identified the motif, and drove the contested biostratigraphic questions to a **documented position** — a written answer, not an assumption carried forward.

WHAT FOLLOWED

A regional framework that could be handed over as a working system — the kind of artifact that keeps a team aligned years after the author has moved on. This is where the KL2 time-depth review later connected back in.

WHAT WAS PROVEN

Integrating sparse, conflicting datasets into a regional model — and naming the gaps rather than papering over them.

PUTERI BASEMENT-1 · MALAY BASIN · 2017

A play that was not in the standing portfolio.

EXPLORATION GEOSCIENTIST · PLAY ORIGINATOR

THE PROBLEM

Basement was not part of the accepted portfolio in this basin — no template, no analogue, no standing prospect. Generating it meant building the case from first principles.

THE EARTH SIGNAL

Integrated aerogravity, seismic and well data supported a pre-salt play concept targeting a fractured basement reservoir. Pre-stack depth migration was required to place the target correctly.

ROLE & DECISION

Originated the play concept and its evaluation. Characterised the specific risks a basement test carries — lithology, charge, top seal — rather than transferring clastic-play assumptions onto a different system.

WHAT FOLLOWED

A first basement test in the trend, and a documented risk characterisation that outlives the well itself. The play existed in the portfolio afterwards because someone built the case for it.

WHAT WAS PROVEN

Generating a new play outside the accepted template — the rarer skill in a mature organisation.

The honest close-out, and the dry-hole corpus.

BUNGA TASBIH-1 · PM328 · MALAY BASIN · 2024

Closing a marginal play on the evidence.

LEAD GEOSCIENTIST · POST-DRILL PLAY ASSESSMENT

THE PROBLEM

The play was marginal, and the expected temptation after a costly well is to extend it on narrative rather than close it on evidence. That temptation is the quiet source of much wasted capital.

THE EARTH SIGNAL

The prospect had been generated from **MBR+ (mass-balance residual) mapping**, with a sequence-stratigraphic framework used to identify a turbidite fairway. Post-drill, that framework had to be held to account.

ROLE & DECISION

Co-authored the well proposal and the G&G authorisation documentation, then led the post-drill assessment that **closed the marginal play** on the evidence rather than extending it.

WHAT FOLLOWED

A disciplined close-out — the kind that protects capital later. In exploration, the willingness to say **stop** is worth more than the willingness to say **drill**, and it is much rarer in a career record.

WHAT WAS PROVEN

Honest portfolio close-outs. The discipline that makes the successes credible.

INHERITED DRY-HOLE CORPUS · SB403 / SB505 / SB412 · SABAH

Converting history into calibrated risk.

SUBSURFACE ASSURANCE · PROSPECT-RISK CALIBRATION

THE PROBLEM

Six inherited dry holes — Solisip, Pekaka, Maligan, Kudat, Balambangan, Kalutan — plus the early-career **Wakid-3** deepwater evaluation, sat in the record as failures with no shared calibration value.

THE EARTH SIGNAL

Auditing wells that did not find hydrocarbons is how a fairway's real risk profile is discovered. Each dry hole is an anchor case for the outboard prospect in the Kinabalu fairway.

ROLE & DECISION

Audited the inherited corpus into **calibrated anchor cases** for outboard prospect risk — converting historical failure into usable decision input.

WHAT FOLLOWED

A risk model grounded in this basin's own history rather than a generic analogue — and a documented instance of treating failure as evidence, not as something to be quietly buried.

WHAT WAS PROVEN

Converting historical failure into downstream decision value.

One section, many answers.

An interpretation is never a finished drawing. It is the current best answer to a set of questions, held open until either the seismic, the well, or an independent dataset says otherwise — and then revised, in writing.

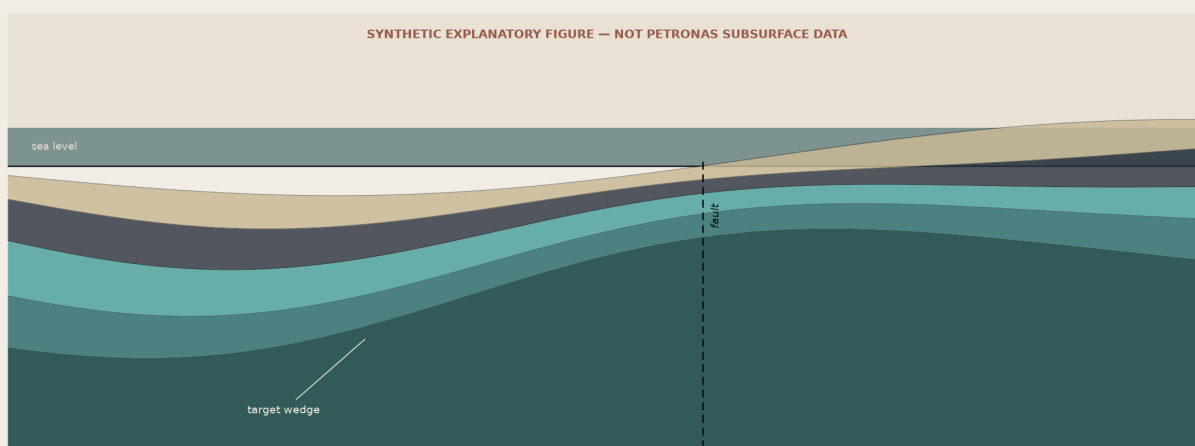


FIGURE 05 — Synthetic explanatory cross-section. **NOT PETRONAS SUBSURFACE DATA**. This is a schematic constructed to show the method — the wedge geometry, the fault relationship, the horizon framework — and it carries nothing from any real prospect, well, or licence.

THE METHOD

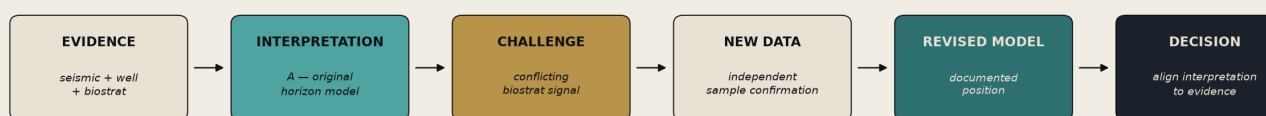
Build the structural framework first. Tie it to the wells. Then ask what the stratigraphy is doing **between** the control points. The inter-well question is the one that decides everything.

THE DISCIPLINE

Competing interpretations are kept alive in the record until one is falsified — not merged into an average, and not lost when a favourite wins. The one that survives should be the one that explains the most with the least special pleading.

How a model actually changes.

The 2026 KL2 time-depth review is the clearest example in the record of something more useful than being right the first time: **knowing when the data has contradicted you, and saying so in writing.**



humility is not the absence of a position — it is the documented path between positions

FIGURE 06 — Interpretation revision path. Sequence modelled on the documented KL2 review (Jun 2026), where an independent biostratigraphic check from the EGRS domain specialist confirmed an earlier correction cascade. Synthetic schematic — no confidential content.

The signal that was easy to miss

In the KL2 well dataset, one well carries a literal **"SYNTHETIC"** label in its header row. A human spots it; an automated ingest pipeline must be built to spot it too, or the correction never happens. A quality-control design problem, not a geology problem.

The deviation that mattered

A deviated well in the same dataset needs a **ray-traced** time-depth correction, not a straight-line vertical projection — a difference of several milliseconds, which in a thin reservoir is the difference between a target and a miss. Details like this decide wells.

The loop that never closes cleanly.

Basin to play to prospect to well to result — and then back to basin, because the result changes what the basin means. The commercial layer attaches to this loop, but it does not replace it: no economic case survives a geology that was never true.

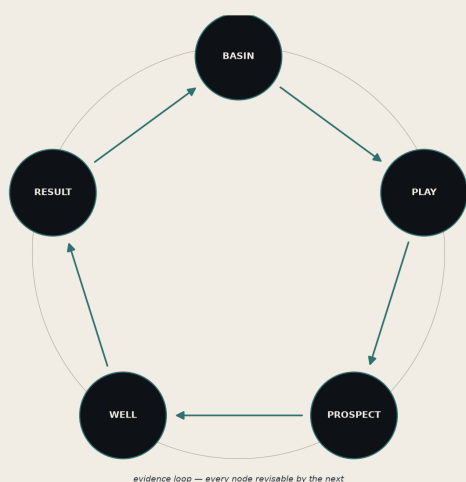


FIGURE 07 — The evidence loop. Every node remains revisable by the next. Synthetic schematic.

- **Risk is not pessimism.** It is the price of the information you do not yet have. A properly risked prospect can be a good decision even when the well comes in dry.
- **Value of information.** The most expensive data is the data you bought too late to change a decision. Sequencing is a technical skill.
- **Reversibility as a design goal.** Prefer the interpretation that can be re-opened. Prefer the decision that can be paused. Prefer the model that declares its own uncertainty rather than hiding it.
- **The result always wins.** The well is the only witness that cannot be argued with. When it disagrees with the model, the model changes — that is not defeat, it is the method working.

NOTE ON ECONOMICS — This page deliberately describes decision *structure*, not monetary outcome. No NPV, IRR, EMV, or reserve figure is stated anywhere in this document. Those figures are not publication-safe, and inventing them would destroy the credibility of everything else.

Expertise is the reasoning, made transferable.

The answer is the smallest part of what a senior geoscientist carries. What matters is whether the reasoning behind it can survive the person leaving the room — and whether the next person can re-open it when the data changes.

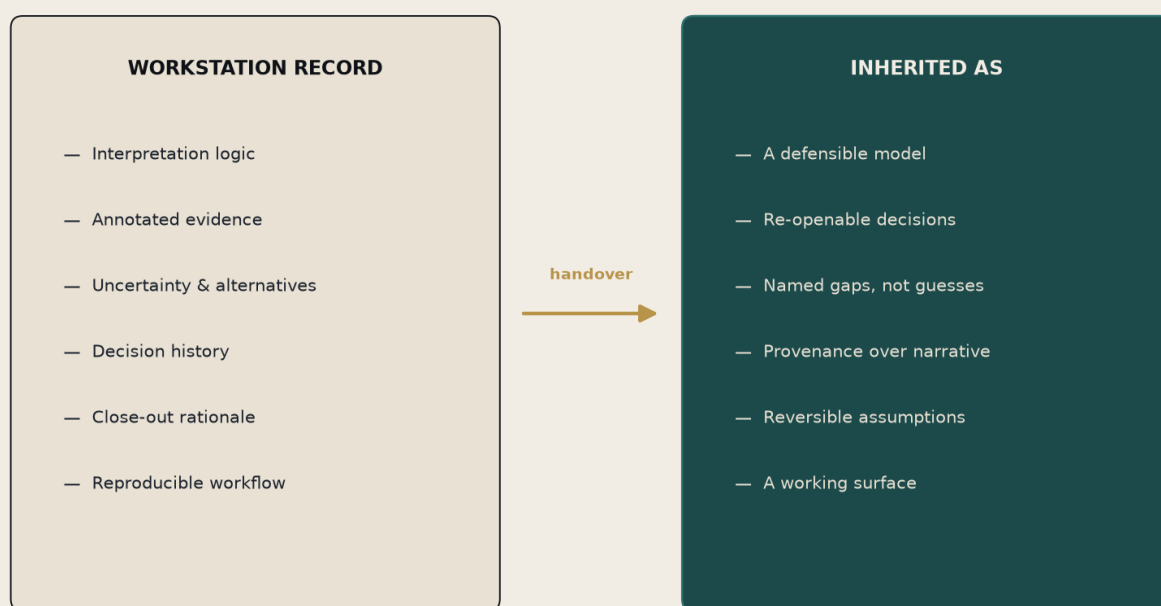


FIGURE 09 — What transfers. The artifact that outlives a workstation is not the map itself but the reasoning that produced it — including the alternatives that were rejected and why. Synthetic schematic.

MENTORING, BOTH DIRECTIONS

Thirteen years of Malay Basin to pass on — and still a great deal to learn from people who read the same data differently. The stated development priority is not more breadth; it is **having interpretations read and tested**, which is the only thing that sharpens a geologist.

HANDOVER AS CRAFT

A handover is done properly when the person receiving the portfolio understands what they are receiving — the model, the caveats, the open questions, the decision history. Not a file dump. A working surface that the next person can stand on.

An amplifier, not a replacement.

This is not a technology advertisement, and it is not the arifOS federation's story. It is one line of personal development: the surface a geologist's discipline has been applied to, over a career.

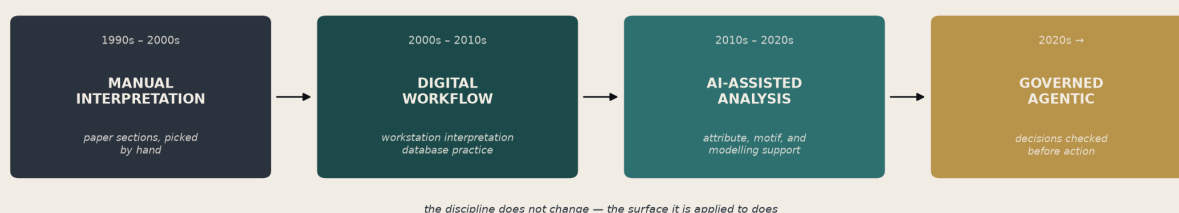


FIGURE 08 — Four surfaces, one discipline. The habit that moves from stage to stage is the same: build the best defensible model, and stay honest about its confidence.

What carries across

Exploration teaches one durable habit: **a model's confidence is not evidence**, and the cheapest error is the one caught before the bit turns. Applied to machine intelligence, that habit becomes a governance layer — an action checked against constraints *before* it runs.

ARIFOS · PUBLISHED · AGPL-3.0 · INDEPENDENT OF EMPLOYER

What does not

AI does not replace geological judgment — it makes the cost of unstructured judgment higher, because output arrives faster and in more volume. The discipline of knowing which model to trust is **more** valuable in an AI-augmented workflow, not less.

POSITION, NOT PROMOTION · AN OPINION HELD AS AN OPINION

arifOS is an independent personal initiative. It is not an employer project, and nothing in this section is affiliated with or derived from PETRONAS work. The only relevant fact is transferable: constraint-first thinking, provenance that survives audit, and decisions that are reversible by default.

The map is not the basin.

The seismic is not the geology.

The model is not the reservoir.

The AI is not the judgment.

Every representation is useful
only while it remains answerable
to reality.

Where **this** is heading.

Not a resignation narrative. Not a departure story. A statement of where the technical work goes next — and what kind of role it is ready for.

DEPTH

Deeper geoscience in one theatre, not surface-level across many. Interpretation that can be **tested**, tested again, and handed over intact. The stated target is the technical ladder, at the principal-geoscientist level — the work of being the reference point a team brings a seismic line to.

SYSTEMS

The governance instinct is not a hobby — it is what happens when a discipline built on uncertainty is applied to any system that makes decisions under incomplete evidence. That includes AI systems, and it will include the next one.

TRANSFER

Making the reasoning transferable is not a soft skill appended to the technical work; it is part of the technical work. A framework that cannot be handed over is not finished.

PROFILE OF THE NEXT ROLE

TECHNICAL SCOPE

Basin-scale subsurface leadership — owning the interpretation end to end, from regional framework to drill-or-drop, with the authority to close a play honestly.

WORKING STYLE

Deep in the data, honest about uncertainty, and willing to build systems that make the reasoning survive the individual. Mentoring both directions.

VALUE DELIVERED

Wells that find what was predicted, plays that were tested when the portfolio said no, and capital protected by close-outs taken on the evidence.

NOTE ON FRAMING — Current documented title is *Executive Geoscientist*. "Principal" appears here only as the named target on the technical ladder (career form, 29 Sept 2026), never as a claim of having held the title.

Every claim, tagged.

This appendix exists so the document can be audited rather than believed. Each headline statement carries a truth class and a source reference.

DOCUMENTED	· career record — CV v2 Sept 2026 · career form — 29 Sept 2026 verbatimio — FORGED, NOT GIVEN · KL2 / Kinabalu — GEOX eureka notes
12 sources · 8 file-backed · hash-listed	· layang-Layang contrast note · PETRONAS institutional context · MSS evidence — form + email · delivery target — people.yaml
SELF-REPORTED	· career aspiration (Arif, unedited, 29 Sept 2026)
1 source · owner testimony · labelled	
SYNTHETIC	· stratigraphic hero · regional schematic map · cross-section mock
figures only — not subsurface data	

every headline claim traces to a source and a truth class

CLAIM AREA	TRUTH CLASS	SOURCE / BASIS
Identity, career arc	DOCUMENTED	CV v2 (Sept 2026); biography <i>Forged, Not Given</i> ; career form (29 Sept 2026, verbatim).
13 years, 2013–2026	DOCUMENTED	CV v2 experience section; PETRONAS staff number 1008142 on the career form.
Lebah Emas-1 · eleven reservoirs	DOCUMENTED	CV v2 selected exploration record. Well name is a published identifier.
Bekantan-1 · shallowest oil tested	DOCUMENTED	CV v2 selected exploration record.
Bunga Tasbih-1 · MBR+ / turbidite	DOCUMENTED	CV v2; ExxonMobil CV (Apr 2026) independently names the same well and method.
Puteri Basement-1 · play originator	DOCUMENTED	CV v2; ExxonMobil CV names the same well and pre-salt concept.
Kinabalu IBS · 88-well database	DOCUMENTED	CV v2; GEOX KL2 / Kinabalu eureka notes (Jun 2026) corroborate the well set.
Inherited dry-hole corpus	DOCUMENTED	CV v2 names Solisip, Pekaka, Maligan, Kudat, Balambangan, Kalutan and Wakid-3.

Claims held back, and why.

An evidence appendix that only lists what supports the story is not an appendix — it is marketing. This page records what was deliberately **not** claimed.

STATEMENT CONSIDERED	DISPOSITION	REASON
"100% exploration success / zero dry wells"	NOT CLAIMED	Self-reported career narrative. The public record itself names dry holes (Wakid-3 and six inherited wells). The claim and the record contradict each other, so neither is published as achievement.
"Discovery leader" / "20% cycle reduction"	NOT CLAIMED	No document in the source set substantiates the numbers. High-bar claims require a source; where a number has no source, the number is dropped, not softened.
Reserve / resource volumes	EXCLUDED	Not publication-safe. No GIIP, STOIP, or recovery figure appears in this document.
Economic outcomes (NPV / IRR / EMV)	EXCLUDED	Commercial figures are not publication-safe and cannot be defended from the source set.
Internal maps, seismic, well logs, coordinates	NEVER REPRODUCED	Confidential by default. All visual content in this document is either synthetic schematic or sourced from public-domain literature.
Title: "Principal Geoscientist"	FRAMED AS TARGET	Documented title is <i>Executive Geoscientist</i> . "Principal" is presented only as the named next-ladder target, never as a held title.
Institutional matters (restructuring, exit process)	PRIVATE	This artifact is a <i>technical</i> career record. Employment-process detail is excluded by design.

CONFIDENTIALITY POSTURE — This document contains no proprietary reservoir, well-test, acreage, coordinate, or economic data. Visual content is synthetic or public-domain. Where an evidence source was private (personal CV drafts, correspondence), it is cited as a source category but its contents are abstracted, not reproduced.

How this was built.

BUILD RECORD

Compiled 2 October 2026, Kuala Lumpur.

Renderer WeasyPrint 70.0 · A4 · embedded fonts.

Figures 10 — all generated programmatically, truth class synthetic-explanatory except where stated.

Source set 12 items — 8 file-backed, hash-listed; 1 owner testimony; 3 synthetic-figure classes.

PUBLICATION LAW APPLIED

Reality > Model > Story >

Aesthetics. Every impressive sentence in this document is paid for by an entry in the evidence appendix. Where the payment could not be made, the sentence was removed — see Appendix II.

WHAT THIS IS NOT

Not a CV. Not a marketing brochure. Not an internal PETRONAS document. It is a **technical career artifact**: evidence, craft, technical memory, human judgment, and a future trajectory, in one place.

The object of the work was never to be certain.

It was to be **honest** about how much could be defended.

Everything else — the wells, the plays, the frameworks — is what that honesty produced.